

Monocular Depth Estimation

Project Introduction

March 27, 2026

Depth Estimation from RGB Images

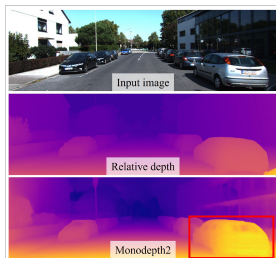
- Goal: Predict depth (distance) from a single RGB image
- Input: RGB image $H \times W \times 3$
- Output: Depth map $H \times W \times 1$



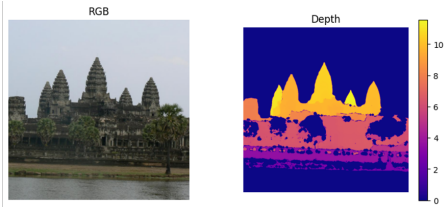
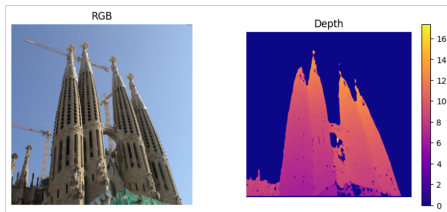
Figure: RGB input, Ground Truth depth

Why is Depth Estimation Important?

- **Robotics & Autonomous Driving**
 - Obstacle avoidance
 - Scene understanding
- **Augmented Reality (AR)**
 - Object placement in real environments
- **Medical Imaging**
 - Spatial understanding from 2D scans



Dataset Examples



Dataset Overview

- Training samples: 22605
- Test samples: 591
- Input: $560 \times 560 \times 3$ (RGB)
- Output: $560 \times 560 \times 1$ (Depth)
- Depth range:
 - Minimum: 0.001
 - Maximum: 80
 - No global distance unit

Evaluation Metric: siRMSE

- Scale-invariant Root Mean Squared Error
- Accounts for relative depth rather than absolute scale

$$\text{siRMSE} = \sqrt{\frac{1}{n} \sum_i d_i^2 - \frac{1}{n^2} \left(\sum_i d_i \right)^2}$$

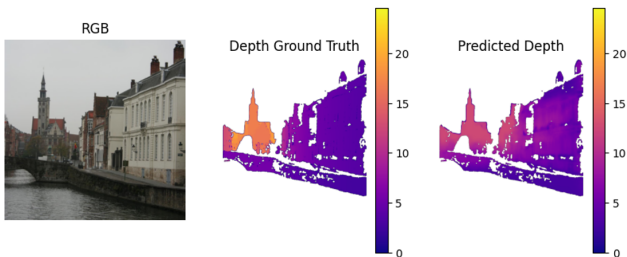
where $d_i = \log y_i - \log y_i^*$

Why siRMSE Matters

- Penalizes incorrect **relative structure**
- Robust to global scaling errors
- Encourages:
 - Correct depth ordering
 - Scene consistency

Valid Pixels

- Ground truth is only available on valid pixels
- Mask zero depth pixels in training/evaluation



How to Start

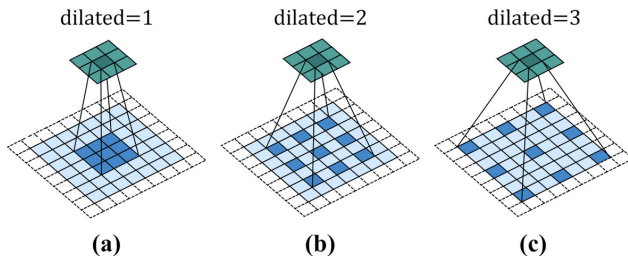
- Input and output have same spatial resolution
- Dense prediction task
- Bottleneck architectures to capture global information
- Preserve local information

Dilated Convolutions

- Increase receptive field without reducing resolution
- Capture global context
- Useful for depth estimation

$$y[i] = \sum_k x[i + r \cdot k] w[k]$$

where r is dilation rate



Summary

- Depth estimation is a dense prediction problem
- siRMSE focuses on relative structure
- U-Net gives you a strong baseline